

Adsorption-Driven Corrosion Inhibition Of Mild Steel By Triclabendazole In Acidic Media: Experimental And Dft Insights

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Abstract

The effects of Triclabendazole on the mild steel against the 1.0 M HCl and 0.5M H₂SO₄ solutions were studied by the methods of **gravimetric**, electrochemical, adsorption, surface charge, and quantum chemical investigations on corrosion inhibition. The loss of weight experiments indicated that the rate of corrosion reduced significantly as the concentration of the inhibitor increased to reach maximum inhibition rates of approximately 96 percent in hydrochloric acid and 93 percent in sulfuric acid. Increased charge transfer resistance, and reduced double-layer **capacitance** were observed in the presence of Triclabendazole using electrochemical impedance spectroscopy, as evidence of the development of a protective adsorbed film. The strong polarization of potentiodynamic polarization proved that Triclabendazole is a mixed **type of** inhibitor in that it inhibits anodic reaction as well as cathodic reactions. The process of inhibition according to adsorption studies followed the Langmuir adsorption isotherm, the negative values of the free energy showed spontaneous adsorption with both physical and chemical interactions. The experimental findings were corroborated with density functional theory electronic, which indicated desirable electronic characteristics and increase in reactivity of protons Triclabendazole species and reactivity in acidic conditions.

Keywords: Corrosion inhibition; Mild steel; Triclabendazole; Electrochemical studies; Density functional theory.

1. INTRODUCTION

Mild steel is common in industry due to its cheap cost and its mechanical ability of being versatile. But, when administered in acidic conditions (pickling, oil-well acidizing, cleaning) mild steel turns out to be corroded easily [1]. Material loss, safety hazards, and huge economic expenses [2] are the results of corrosion in acid media. Conventionally used inhibitors of corrosion (chromates, heavy-metal salts) are efficient but in most cases toxic and ecological issues[3]. The research community, therefore, tries to find green alternatives, which are effective, economic and low **toxicity** [4]. Pharmaceutical compounds, such as expired drugs, are becoming good potential green **inhibitors** [5]. Active sites are blocked by many drug molecules, heteroatoms (N, O, S) and aromatic rings and functional groups that polarize on steel surfaces, slows down anodic and cathodic reactions and reduces corrosion current [7]. Repurposing expired drugs utilizes waste material, which decreases the disposal burden and reduces the cost of production [8].

The electrochemical and surface method allows the determination of the adsorption isotherms as well as the properties of the adsorption films [9]. Computational techniques (DFT, molecular dynamics) are starting to be used to supplement experiments to understand electronic causes of inhibition [10]. An increasing literature records high inhibition efficiencies (>80-95%) with numerous expired drugs in HCl or H₂SO₄ media [5,11]. These are examples of expired antibiotics and antifungals formed on steel in the 1M HCl as stable adsorbed layers [12]. Physisorption and chemisorption processes have been described, either with Langmuir or Temkin isotherms, both depending on the situation [13]. Mechanistic knowledge connects inhibition to: lone-pair donation to iron, π -d interactions of aromatic rings, and film formation that is change of resistance of charge transfer [14]. Differences between physical and chemical adsorption are achieved through temperature studies and analysis of activation energy [15]. In particular systems adsorption and inhibition may be further optimized by the use of synergistic additives (halide

ions, surfactants) [16]. Lifecycle and environmental factors are of significance: on the one hand, expired drugs help to decrease the amount of pharmaceutical waste, and their ecotoxicology has to be evaluated before the implementation on large-scale basis [17]. The field application should be based on regulatory and safety reasons (handling, leachability, by-products) [18].

Some of the research issues are the standardisation of the test condition, transfer of laboratory findings to the industrial environment, and the long-term integrity of films of inhibitors [19]. There are possibilities of combining green formulation

approaches and computational screening of the expired drugs to obtain the best corrosion inhibitors within a short period of time [20]. The expired drugs offer a cost-efficient, sustainable method of controlling corrosion of mild steel in acidic environments. Before it can be widely used, careful experimental validation, mechanistic study and evaluation of environmental effects should be done. Therefore, an effort is made to apply expired drug Triclabendazole, which is usually an antifungal drug [21] as corrosion inhibitor to manage corrosion in mild steel in acid media. The Figure.1 provides the structure of drug molecule.

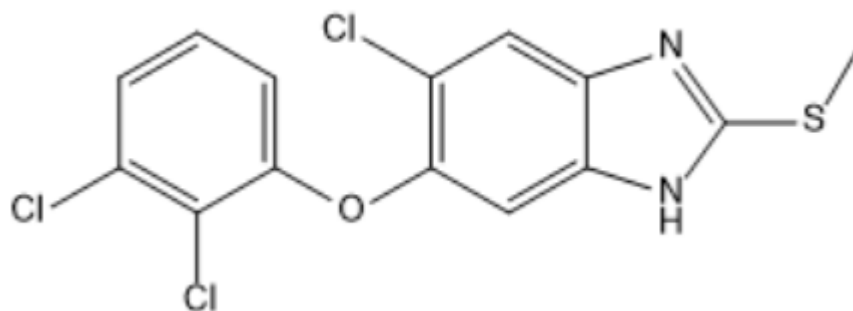


Figure.1 Structure of Triclabendazole.

2. EXPERIMENTS

2.1 COMPOSITION OF THE TEST MATERIAL

The mild steel material used in this study is composed of the following elemental composition. 0.086 C 0.039 Mn 0.027 P 0.022 S and the residual Fe found as a result of the process.

2.2 PREPARATION OF THE SPECIMEN

The mild steel specimen to be studied by gravimetric analysis was of dimensions 2.5 x 1 x 0.1 cm. Mechanical polishing of the specimens was done using emery sheets of various grades such as 220, 400, 600, 800 and 1200 and the specimens were washed with double distilled water. The specimens were analyzed, and then the acetone was used to de-grease them, dried and stored under a desiccator. The electrochemical studies were carried out using Teflon coated mild steel cylinder-shaped rod in the exposed area of 1cm². Test electrode was also cleaned after each run as stated above. All of the solutions were made with the help of the double distilled water.

2.3 PREPARATION OF ACID SOLUTIONS

The 1000 ml standard measuring flask of analytical level of the concentrated acids (HCl and H₂SO₄) was diluted with double distilled water and the normality was found by titration with the standard sodium carbonate solution preparation of the inhibitor solution.

2.4 PREPARATION OF INHIBITOR SOLUTIONS

Triclabendazole was taken as a stock from available commercial tablets and determined following a procedure reported by R. Mruthunjaya Rao et.al [21] followed by addition of 3% DMSO solvent to make it soluble. Dilution of the stock solution produced the concentrations of Triclabendazole required.

2.5 WEIGHTLOSS MEASUREMENTS

Weight loss measurement was conducted with ASTM procedure to measure the effectiveness of corrosion and corrosion inhibitor [22]. The mild steel samples that had been prepared in triplicate were thoroughly immersed in 100ml of 1M HCl solutions and 0.5M H₂SO₄ solutions with various concentrations of the inhibitor in 30 ± 1° C temperature after which they were washed, dried and weighed. The mean of the weight loss was then used in the calculation of the corrosion rate and corrosion [23] efficiency was obtained as follows.

$$IE(\%) = \frac{W_{BS} - W_{IS}}{W_{BS}} \times 100$$

Where, IE is the efficiency of the inhibitor, WBS is the weight loss in an inhibited solution and WIS is that of weight loss in an inhibited solution.

2.6 Electrochemical impedance spectroscopy

Through the potentiostat / Galvanostat (Zahner Zennium XC superior Electrochemical Workstation), electrochemical studies have been

conducted. In three electrode system, the working electrode labelled Teflon coated mild steel rod with a 1 cm² open area was used, a saturated calomel electrode was used as the reference and platinum mesh was used as counter electrode. Measurement of the EIS was conducted another time in the range of 106 to 10⁻² Hz at the open circuit potential under the perturbation of 10 mV by the applied AC sinusoid signal after 30 minutes immersion in the acid media. The values of the inhibition efficiencies were determined using the values of the charge transfer resistance (R_{ct}) as obtained in the equivalent circuit fits as depicted below [24].

$$\% IE = \frac{R'_{ct} - R_{ct}}{R'_{ct}} \times 100$$

Where R'_{ct} is the charge transfer resistance in the inhibited solution, R_{ct} is the charge transfer resistance in the uninhibited solution.

2.7 Polarization measurements

The potentiodynamic polarization curves were taken at a more negative potential than E_{ocp} up to more positive potential than E_{ocp} using E_{corr} at 1.67 mV/s. The values of β_a and β_c of the Tafel constants have been determined by plotting of E Vs $\log i$. The anodic and cathodic polarization curve were drawn and i_{corr} and E_{corr} have been calculated using the intersection point of the two tangents. The efficiencies of inhibition were obtained by use of (3) equation [25].

$$\% IE = \frac{i_{corr} - i'_{corr}}{i_{corr}} \times 100$$

Where i_{corr} is the corrosion current density in uninhibited solution and i'_{corr} is the corrosion current density of inhibited solution.

2.8 Measurement of the potential of zero charge (PZC)

The adsorption can be a result of the attraction forces caused by the electric charges between the adsorbed species and ionic charges on the metal and the electric charge on the metal at the metal - solution contact point. The potential of a metal charge in solution with respect to the zero-charge potential measures the charge of a given metal charge. This potential compared with the zero-charge potential, commonly known as the ϕ -potential, is more noteworthy with regard to adsorption than is the potential on the hydrogen scale, and actually the behaviors of these two potentials can be distinct. To determine the effect of the charge on the organic adsorption, Antropov came up with the potential difference between the corrosion potential and the PZC of a corroding metal. The more positive the potential of a metallic surface, the more the adsorption of anion is promoted, and the more negative the ph-potential, the more is the adsorption of cations.

The Impedance response of the corroding electrode at 100Hz at an AC amplitude of 10mV value of Cdl at various applied DC potentials and pzc were obtained by plotting the difference capacitance values divided by the applied DC potential.

The adsorption of the inhibitor on the metal surface is largely determined by the charge that is present on the surface. The E_r sign is the formula below [Popova 2003].

$$E_r = E_{ocp} - E_{PZC}$$

Where,

E_{PZC} = Potential of zero charge

E_r = Antropov's rational corrosion potential

E_{OCP} = Open circuit potential

2.9 Quantum Chemical Studies

In the present study, density functional theory (DFT) calculations were employed to investigate the electronic properties of the inhibitor molecule and its protonated forms, which are expected to exist in acidic environments. Acidic media promote protonation of heteroatoms such as nitrogen, sulfur, or oxygen, significantly altering charge distribution, molecular reactivity, and adsorption behaviour. Therefore, considering both neutral and protonated species is crucial for accurately correlating theoretical results with experimental corrosion inhibition performance.

Quantum chemical parameters such as the energies of the highest occupied molecular orbital (E_{HOMO}) and lowest unoccupied molecular orbital (E_{LUMO}), energy gap (ΔE), ionization potential, electron affinity, global hardness, softness, electronegativity, electrophilicity index, and fraction of electrons transferred were calculated. These descriptors are widely recognized as reliable indicators of an inhibitor's ability to donate or accept electrons, interact with vacant d-orbitals of metal atoms, and form stable adsorbed layers on the metal surface.

Geometrical optimization of the inhibitor molecules was carried out to obtain the most stable molecular conformations, ensuring that all calculated electronic properties correspond to minimum-energy structures. Frontier molecular orbital analysis was used to identify active adsorption centers and to visualize regions of high electron density responsible for metal-inhibitor interactions. The correlation between calculated quantum chemical parameters and experimentally observed inhibition efficiencies enables a deeper understanding of the adsorption mechanism and validates the role of electronic structure in corrosion protection. Thus, quantum chemical studies serve as a powerful theoretical tool to support experimental findings, elucidate the inhibition mechanism at the atomic level, and predict the effectiveness of corrosion inhibitors in aggressive acidic environments.

3. Results and Discussion

3.1 Weight loss measurement

Table.1 presents the measurement of the weight loss of mild steel when corroding in in 1.0 M HCl as well

as in 0.5 M H_2SO_4 acid solution in the presence of Triclabendazole.

Table.1 Findings of weight loss corrosion of mild steel in 1.0M HCl and in 0.5M H_2SO_4 acid solutions.

Medium	Inhibitor concentration (mM)	Corrosion rate (mpy)	% IE
1.0 M HCl	Blank	1267	-
	0.1	910.9	28.1
	0.5	642.4	49.3
	1.0	401.6	68.3
	2.0	337.0	73.4
	5.0	48.15	96.2
	10.0	49.41	96.1
0.5 M H_2SO_4	Blank	1488	-
	0.1	1123	24.5
	0.5	903.2	39.3
	1.0	697.8	53.1
	2.0	465.7	68.7
	5.0	99.69	93.3
	10.0	102.7	93.1

As can be seen in Table 1, the rate of corrosion of mild steel reduces significantly with the acid mixture content of Triclabendazole, with the corresponding rise in the efficiency of inhibition (percentage inhibition). The rate of corrosion in 1.0 M HCl is reduced in the presence of 1267 mpy (blank), 48.15 mpy at 5.0 mM with a maximum inhibition efficiency of 96.2%. The same tendency is traced in the case of 0.5 M H_2SO_4 , the rate of corrosion is reduced as 1488 mpy to 99.69 mpy and the effectiveness of the inhibitors is 93.3%. This is possible because the higher the concentration of the inhibitor, the high the adsorption of the inhibitor molecules on the surface of the mild steel and the subsequent coverage of the surface and consequently covers the active corrosion sites [27].

Triclabendazole has greater inhibition ability in 1.0 M HCl than in 0.5 M H_2SO_4 at all concentrations studied. This contrast is usually given with respect to organic inhibitors and is dependent on the nature of aggressive anions in the solution. Chloride ions have been reported to favour the adsorption of organic inhibitors by adjoining ion pairs with protonated molecules of the inhibitor consequently enhancing greater adsorption to the steel surface [28]. Conversely, the sulfate ions are more hydrated and have greater ionic size hence limited in their capacity to help in the adsorption of the inhibitors which quells relatively lower efficiency in inhibition [29]. The adsorption on the mild steel surface is the most important factor that dictates the corrosion inhibition behaviour of Triclabendazole. Based on the nitrogen and sulfur heteroatoms and π -electron-rich rings of aromatic ring, triclabendazole has the active adsorption centers. The adsorption can take place by means of a mixture of:

Electrostatic attraction (physisorption) of protonated molecules of the inhibitors upon the negatively charged steel substrate in acidic media [26].

Bonding (an analogous process called chemisorption) by the donation of lone pair electrons by heteroatoms to vacant d-orbitals on iron atoms [30].

Interactions of ring aromatic rings with the metal surface through π -d orbital interactions, which contribute to the stability of the film [29].

A protective layer formed by the adsorbed molecules of the inhibitor prevents both the anodic reaction of iron dissolution and cathodic reaction of hydrogen evolution and as such, the affinity of the inhibitors optimally rises to 5.0 mM after which the rate is not enhanced any more [31]. The insignificance in the inhibition efficiency at 10.0 mM of inhibitors in both acid media, can be explained by a partially desorbed or re-oriented inhibitor molecules at higher concentrations which causes destabilization of the protecting film [28,30]. To the extent that this phenomenon represents, it is apparent that the optimum concentration of the corrosion inhibitor under the explored conditions is 5.0 mM.

3.2 Electrochemical Impedance Spectroscopy

Electrochemical impedance spectroscopy (EIS) is currently a common method of assessing the corrosion processes, as well as inhibitor behaviour, at the electrolyte-metal interface since it enables the separation of various electrochemical processes that occur over a broad frequency range [32]. In the case of mild steel in hostile conditions, the Nyquist plots usually show a single depressed capacitive semicircle with this showing that the corrosion reaction is largely controlled by the charge transfer reactions [33]. Figure. 2 and 3 illustrates the experimental

Nyquist plots. By measuring the experimental curves and conforming them to the equivalent circuit as shown in Figure.4 the electrochemical impedance parameters were obtained by fitting. The lack of an ideal semicircle is explained by the roughness of surfaces, non-homogeneity and non-uniform adsorption of molecules of an inhibitor that requires a constant phase element (CPE) rather than an ideal capacitor in the equivalent circuit modelling [34]. An equivalent circuit often used to model the impedance response so as to fit inclusion of solution resistance (R_s), charge transfer resistance (R_{ct}), and CPE as shown Figure 4.

A positive variation in R_{ct} indicates an improved gas exchange across a metal-solution interface which is equivalent to a reduction in rate of corrosion, and alterations in CPE parameter give indications on the surface coverage and the behaviour of inhibitor adsorption [35].

Figure 2 of the Nyquist plots of mild steel corrosion in 1.0 M HCl solution depict that the Triclabendazole does not change the underlying corrosion process, but inhibits the charge transfer process [36]. The semicircle diameter continues to gradually increase with concentration of the inhibitor because of improved corrosion resistance. Charge transfer resistance (R_{ct}) of the uninhibited solution is $20.2 \Omega \text{ cm}^2$ but at 5.0 mM Triclabendazole, this decreases to $517.9 \Omega \text{ cm}^2$. This high increment of R_{ct} proves excellent inhibitory effect of Triclabendazole by adsorption to the mild steel surface [37]. In parallel, the related inhibition efficacy is 96.1, which implies that a high-protective film is formed as a result of a decrease in the CPE admittance (Y_0) with the increase in the concentration of the inhibitor [34]. The n values that tend to nearness unity mean that surface homogeneity and native capabilities are improved since the effectiveness of inhibitor adsorption is achieved [35].

Depressed capacitive semicircles are also present in Nyquist plots provided in solution 0.5 M H_2SO_4 (Figure 3), which are typical of charge transfer controlled corrosion [32]. The smaller value of R_{ct} of the blank solution $14.2 \Omega \text{ cm}^2$ means that sulfuric acid is being more aggressive on mild steel than the hydrochloric acid [36]. As, the concentration of the Triclabendazole went up, the R_{ct} value was rising progressively, up to a value of $191.9 \Omega \text{ cm}^2$ and an increase in the inhibition efficiency to 92.6%. This affirms that Triclabendazole molecules can effectively inhibit corrosion by adsorbents onto the steel [37]. The reason could be, however, that there is a slight decrease in the inhibition efficiency relative to the HCl medium that could be caused by competitive adsorption of the sulfate ions which decrease the inhibitor adsorption degree and contribute to the raised surface protection and reduced interfacial capacitance [34]. The slightly lower inhibition capability could also be explained by the fact that the sulfate ions are technically competitive to occupy this adsorption surface that in turn lowers the degree of inhibitor adsorption [35]. The resulting decrease in Y_0 and increment in n further support

The results provided by the impedance show clearly that Triclabendazole prevents mild steel corrosion in both acidic systems through an increase of charge transfer resistance, and reduction of double-layer capacitance due to surface adsorption [36]. The inhibitor shows superior performance under an acidic environment HCl than in acidic H_2SO_4 , which can be explained by further stronger electrostatic interaction of the protonated inhibitor molecules with the metal surface and chloride sites of active corrosion control [37]. The fact that the inhibitor is working well in HCl medium when compared to H_2SO_4 could be attributed to the fact that the inhibitor strongly interacts with the metal surface electrostatically and chromosomes of active corrosion control are inhibited by the inhibitor [32].

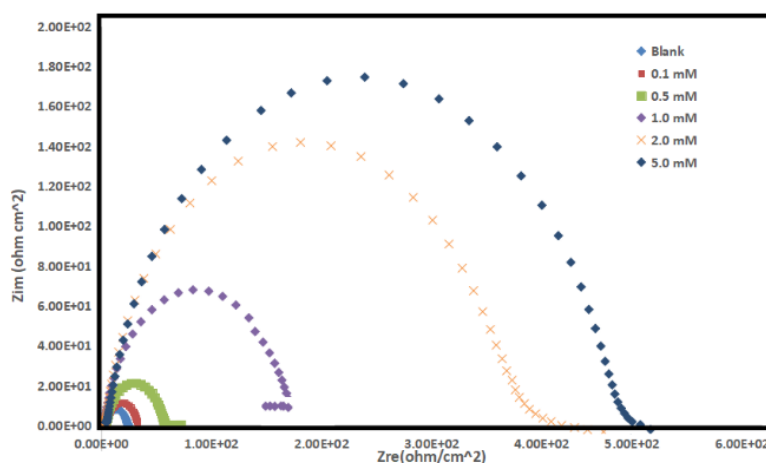


Figure.2 Experimental Nyquist curves of corrosion of mild steel without the presence and the presence of 1.0M HCl solution of triclabendazole.

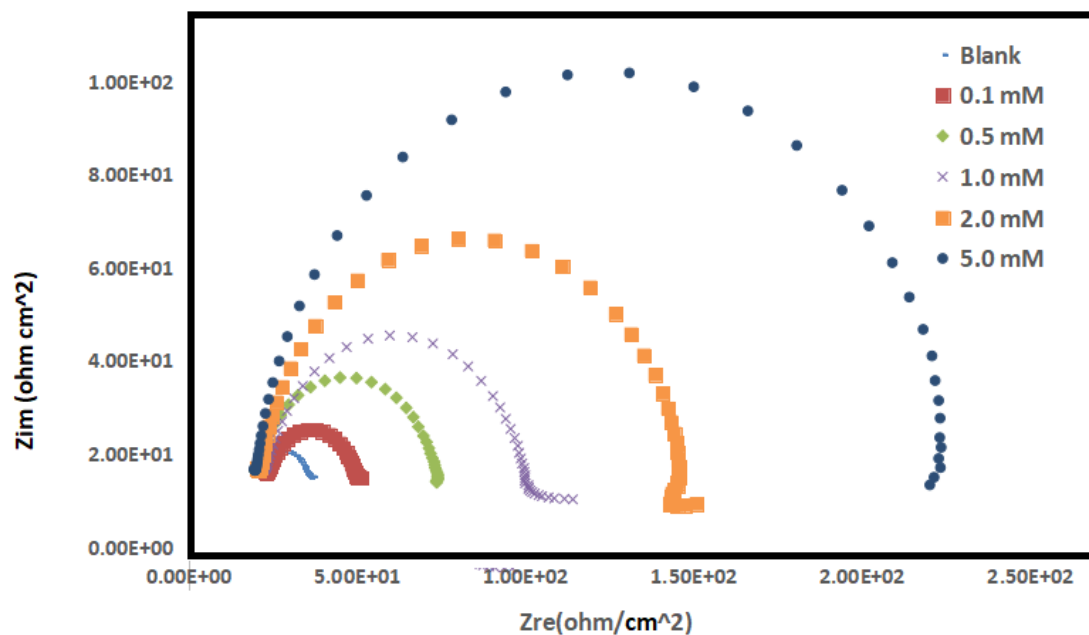


Figure.3 Experimental Nyquist plots of the mild steel corrosion when no single one is present and when they are present Triclabendazole in 0.5 M H_2SO_4 .

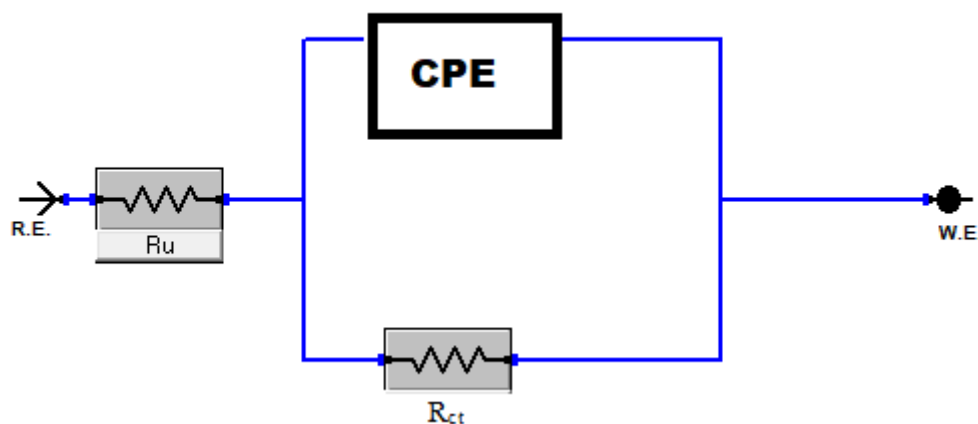


Figure.4 Equivalent circuit to extract electrochemical impedance parameter of an experiment plot.

Table.2 Electrochemical impedance characteristics of corrosion of mild steel in 0.1M HCl and in 0.5 M H_2SO_4 acidic solution.

Medium	Inhibitor concentration (mM)	R_{ct} ($\Omega \text{ cm}^2$)	Y_0 ($10^{-6}, \Omega^{-1} \text{ cm}^{-2}$)	n	% IE
1.0 M HCl	Blank	20.2	82.03	0.882	-
	0.1	29.7	76.22	0.836	32.1
	0.5	53.0	72.09	0.871	61.9
	1.0	169.7	61.03	0.901	88.1
	2.0	297.1	49.83	0.912	93.2
	5.0	517.9	41.08	0.905	96.1
0.5 M H_2SO_4	Blank	14.2	110.2	0.873	-
	0.1	25.2	98.93	0.866	43.7
	0.5	52.8	90.64	0.878	73.1
	1.0	80.2	81.52	0.863	82.3
	2.0	104.4	76.65	0.896	86.4
	5.0	191.9	64.80	0.931	92.6

3.3 Potentiodynamic Polarization Studies

Potentiodynamic polarization is used extensively in order to measure corrosion kinetics, as well as to study the mode of action of corrosion inhibitors by studying anodic and cathodic reactions independently [38]. The method gives important electrochemical values at corrosion potential (E_{corr}), corrosion current density (I_{corr}) as well as anodic (β_a) and cathodic (β_c) Tafel slopes that are directly proportional to the rate of corrosion and mechanism

[39]. These data are given in Table.3. The change in I_{corr} with an inhibitor shows that the corrosion rate is lowered and a change in E_{corr} and alteration in Tafel slopes show whether the inhibitor acts mostly on anodic or cathodic corrosive systems or both electrochemical processes [40]. When the displacement between E_{corr} of the inhibitor and blank is not more than ± 85 mV, the inhibitor is mostly termed as mixed-type inhibitor [41].

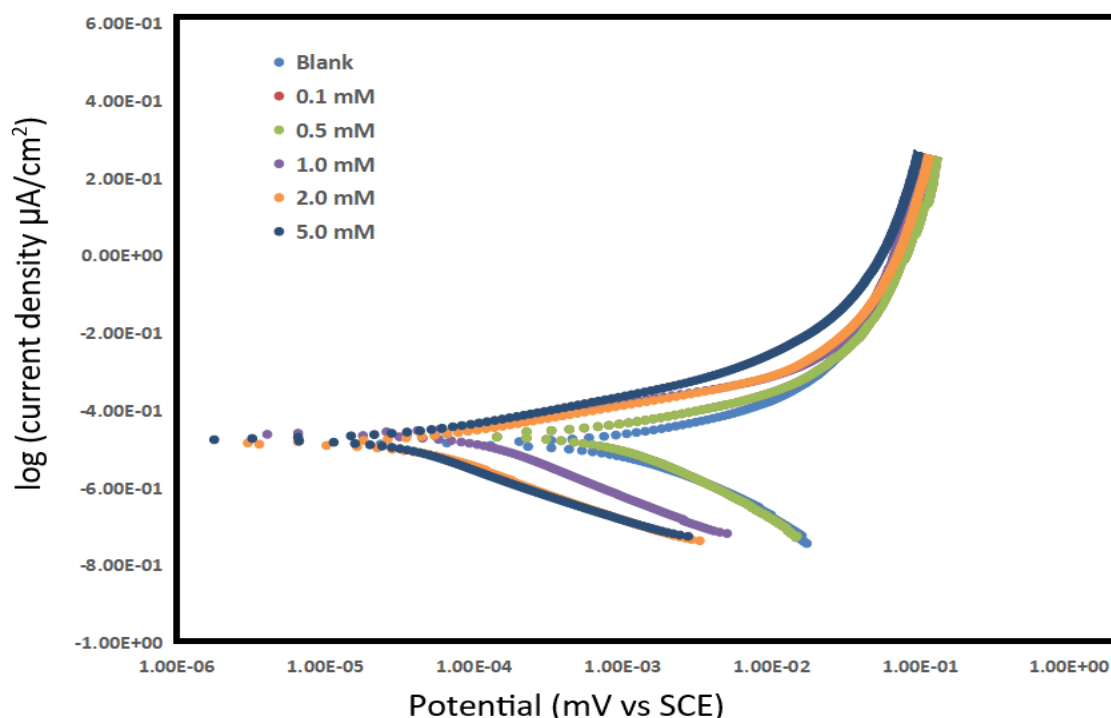


Figure.5 Tafel plots of corrosion of mild steel during the absence and the presence of Triclabendazole in 1.0 M HCl solution.

Figure 5 is a potentiodynamic polarization curve of mild steel in 1.0 M HCl that indicates an anodic and cathodic branches in the absence and presence of Triclabendazole, which means that the reactions of dissolution of steel and the evolution of hydrogen are active [38]. On adding the Triclabendazole, the corrosion current density (I_{corr}) changes to a value of $73.01 \mu\text{A cm}^{-2}$ at 5.0 mM (corresponding to an inhibition efficiency of 95.7%). The large change in anodic (β_a) and cathodic (β_c) Tafel slopes in the presence of the inhibitor confirms the potent inhibitive effect of Triclabendazole by adsorbing onto the mild steel surface [42]. Both the anodic (β_a) and cathodic (β_c) Tafel slopes vary in the presence of the inhibitor and this indicates that Triclabendazole influences the reaction in both dissolution of metals and hydrogen evolution reaction [40]. The change of E_{corr} values is not substantial and it is not more than ± 30 mV which demonstrates that Triclabendazole is a mixed-type inhibitor in hydrochloric acid [41].

Figure 6 displays Tafel polarization curves of mild steel in 0.5 M H_2SO_4 whereby the I_{corr} values as higher than that of HCl are reflected demonstrating that sulfuric acid is more aggressive towards mild steel [43]. When Triclabendazole is added it can be seen that the inhibition efficiency is 91.6% and the I_{corr} values have decreased significantly to $178.8 \mu\text{A cm}^{-2}$ at a concentration of 5.0 mM. Even though the inhibitor is also very effective, the slightly reduced efficiencies compared to HCl imply that the sulfate ions are adsorbed competitively by the Triclabendazole, hence weakening the inhibitor surface coverage [44]. The changes in the values of β_a and β_c with the concentration of the inhibitor means that the Triclabendazole inhibits the anodic reaction as well as the cathodic reaction involving hydrogen evolution [40]. The shifts in the corrosion potential are weak and intermittent, which also proves the presence of the mixed type of an inhibition behaviour in the sulfuric acid medium [41].

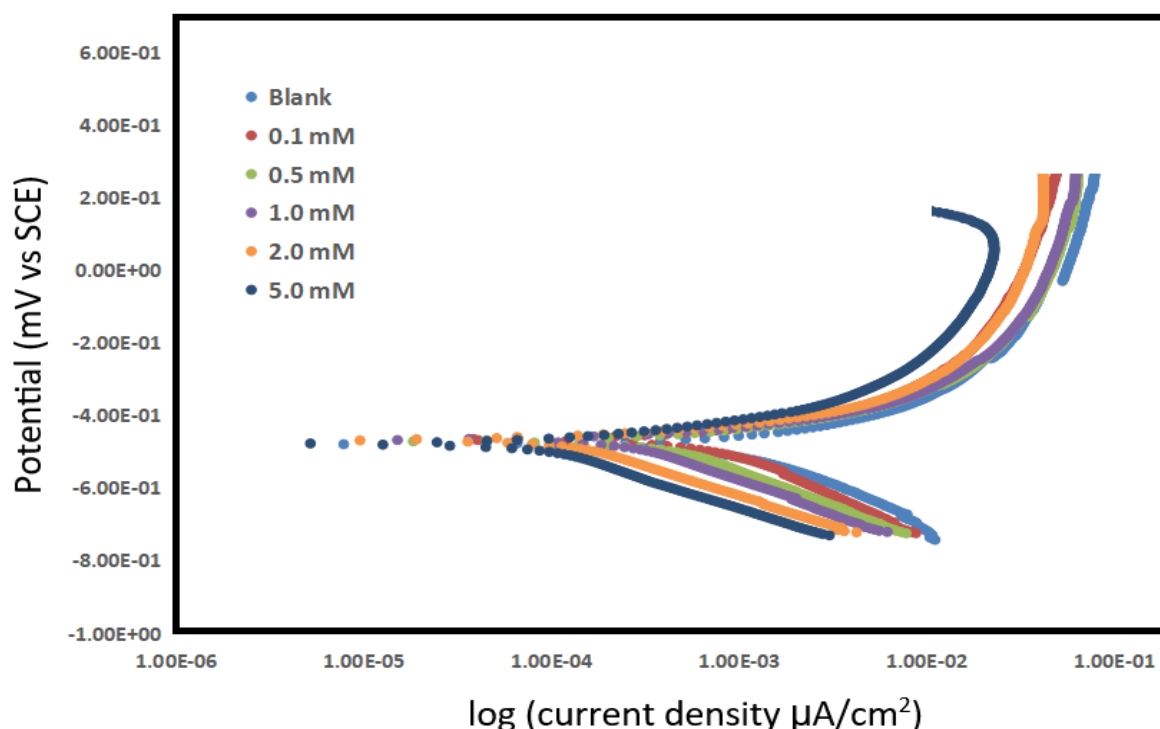


Figure.6 Tafel curves of mild steel corrosion with and without Triclabendazole in the presence of 0.5M H₂SO₄.

The Polarization observed in the two acidic media indicates that Triclabendazole considerably lowers the rate of corrosion of mild steel, mainly by lowering the density of corrosion current [38]. The concomitant change in anodic and cathodic Tafel slopes suggests that protonated inhibitor molecules adsorb to a steel surface covered with chloride active sites in both electrochemical reactions [42]. The

enhanced inhibition of 1.0 M HCl over 0.5 M H₂SO₄ could be explained by enhanced electrostatic interaction of the protonated molecules of an inhibitor with the chloride-covered steel surface, when compared to recognition with sulfate ions [44]. These results are quite consistent with impedance and weight loss literature

Table.3 Potentiodynamic polarization parameters of corrosion of mild steel in 1.0 M HCl solution and 0.5 M H₂SO₄ acid solution in the presence and absence of Triclabendazole.

Medium	Inhibitor concentration (mM)	β_a	β_c	I_{Corr} ($\mu A\ cm^{-2}$)	E_{Corr}	Inhibition efficiency (IE%)
1.0 M HCl	Blank	176	248	1698	-491	-
	0.1	160	251	1219	-467	28.2
	0.5	117	183	672.4	-465	60.4
	1.0	97	178	213.9	-493	87.4
	2.0	87	161	127.4	-485	92.5
	5.0	91	159	73.01	-478	95.7
0.5 M H ₂ SO ₄	Blank	217	381	2129	-482	-
	0.1	214	411	1265	-471	40.6
	0.5	158	321	606.7	-474	71.5
	1.0	120	230	387.4	-468	81.8
	2.0	126	241	300.2	-471	85.9
	5.0	133	272	178.8	-482	91.6

3.4 Adsorption isotherm

The inhibition of metal corrosion by organic compounds is primarily governed by their adsorption onto the metal surface, forming a protective barrier that isolates the metal from the corrosive environment [45]. Adsorption isotherms are used to

describe the relationship between the degree of surface coverage (θ) of an inhibitor and its concentration in solution, providing valuable information about the nature of inhibitor-metal interactions [46].

Among various adsorption models, the Langmuir adsorption isotherm is most commonly applied in corrosion studies. It assumes monolayer adsorption on a homogeneous surface with no interaction between adsorbed species [47]. The Langmuir isotherm is mathematically expressed as:

$$\frac{C}{\theta} = \frac{1}{K_{ads}} + C$$

where C is the inhibitor concentration, θ is the surface coverage, and K_{ads} is the adsorption equilibrium constant [48]. The adsorption isotherm plots for Triclabendazole in 1.0 M HCl and 0.5 M H₂SO₄ (Figure 7) show a linear relationship between C/θ and C with slopes close to unity, indicating that the adsorption of Triclabendazole on the mild steel

surface follows the Langmuir adsorption isotherm [49].

This linearity suggests that Triclabendazole molecules adsorb as a single molecular layer on the steel surface without significant lateral interaction between adsorbed species. The validity of the Langmuir model confirms uniform surface coverage and strong affinity of the inhibitor towards the mild steel surface in both acidic environments [50]. The adsorption equilibrium constant (K_{ads}) values calculated from the Langmuir plots are 5309 L mol⁻¹ for 1.0M HCl and 4957 L mol⁻¹ for 0.5 M H₂SO₄ (Table 4). These relatively high values indicate strong adsorption of Triclabendazole molecules on the mild steel surface [46].

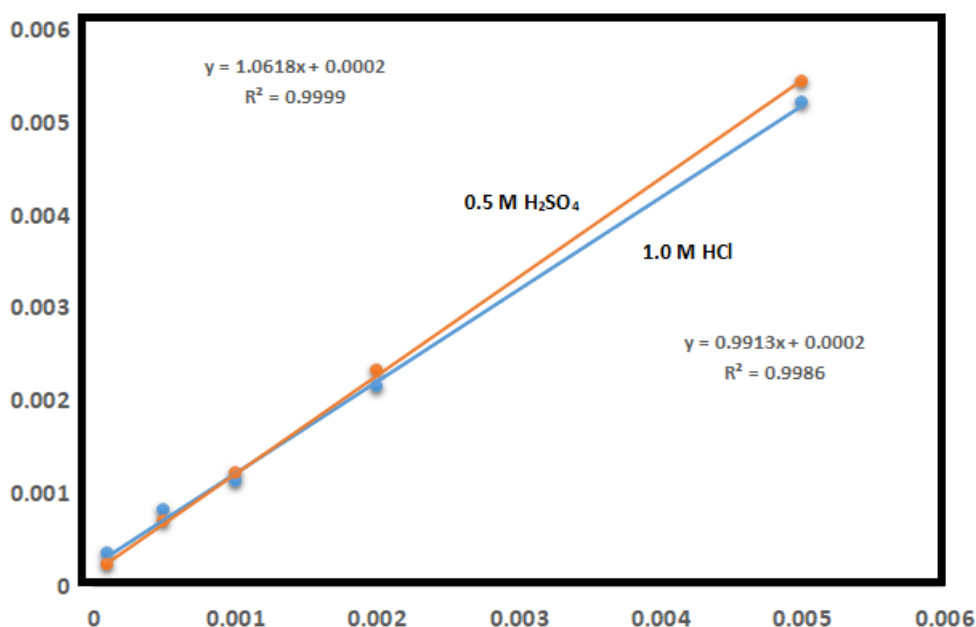


Figure.7 Adsorption isotherm for adsorption of Triclabendazole onto mild steel surface in 1.0 M HCl and in 0.5 M H₂SO₄ acid solutions

The standard free energy of adsorption (ΔG°_{ads}) values were calculated using the relation:

$$\Delta G^{\circ}_{ads} = -RT \ln (55.5 K_{ads})$$

The negative ΔG°_{ads} values of -21.61 kJ mol⁻¹ (HCl) and -21.44 kJ mol⁻¹ (H₂SO₄) indicate that the adsorption process is spontaneous in both acidic media [47]. The magnitude of ΔG°_{ads} values, which lies between -20 and -40 kJ mol⁻¹, suggests that adsorption occurs through a mixed mechanism involving both physical and chemical adsorption, with a dominant contribution from electrostatic interactions [51]. A slightly higher K_{ads} and more negative ΔG°_{ads} value observed in 1.0 M HCl

compared to 0.5 M H₂SO₄ indicate stronger adsorption of Triclabendazole in hydrochloric acid medium [52]. This behaviour can be attributed to enhanced electrostatic attraction between protonated Triclabendazole molecules and the chloride-adsorbed steel surface. In contrast, sulfate ions compete more strongly for adsorption sites, slightly reducing inhibitor adsorption efficiency in sulfuric acid [53].

These findings are consistent with electrochemical impedance and polarization results, which showed superior inhibition performance of Triclabendazole in HCl medium.

Table.4 Thermodynamic Parameters for corrosion of mild steel in 1.0 M HCl and 0.5 M H₂SO₄ in the presence of Triclabendazole

Medium	ΔG (kJmol ⁻¹)	K _{ads}
1.0 M HCl	-21.61	5309
0.5 M H ₂ SO ₄	-21.44	4957

3.5 Potential of Zero Charge

The electrochemical **behaviour** of metals in corrosive environments is strongly influenced by the electrical double layer (EDL) formed at the metal–solution interface. The excess surface charge on a metal surface arises due to the imbalance between electronic charge on the metal and ionic charge in the solution [54]. This surface charge plays a crucial role in determining the adsorption of ions and inhibitor molecules, thereby affecting corrosion kinetics [55]. **The** potential of zero charge (E_{PZC}) is defined as the potential at which the net charge on the metal surface is zero. When the open circuit potential (E_{OCP}) is more positive than E_{PZC} , the metal surface carries a positive excess charge, whereas a negative charge exists when E_{OCP} is more negative than E_{PZC} [56]. The magnitude and sign of the excess surface charge provide valuable insight into inhibitor adsorption mechanisms and electrostatic interactions at the interface [57].

For mild steel in 1.0 M HCl without inhibitor, the E_{OCP} value (−490 mV/SCE) is more positive than the E_{PZC} (−532 mV/SCE), resulting in a positive excess surface charge of +42 mV. This indicates that the steel surface is positively charged in the uninhibited acidic medium, **favouring** the adsorption of aggressive chloride ions, which accelerates the corrosion process [58]. Upon addition of 5.0 mM Triclabendazole, the E_{PZC} shifts positively to −502 mV/SCE, while E_{OCP} becomes −482 mV/SCE.

Consequently, the excess surface charge decreases significantly to +20 mV. This reduction in surface charge indicates that Triclabendazole molecules adsorb onto the mild steel surface and partially neutralize the surface charge, leading to a more stable interfacial structure [59]. **The** decrease in excess surface charge suggests the formation of a protective inhibitor film, which reduces electrostatic attraction between the metal surface and aggressive chloride ions, thereby inhibiting corrosion [60].

In 0.5 M H₂SO₄, the blank solution shows E_{OCP} (−484 mV/SCE) slightly more positive than E_{PZC} (−510 mV/SCE), resulting in a positive excess surface charge of +26 mV. This lower surface charge compared to HCl indicates differences in interfacial ion adsorption **behaviour**, as sulfate ions are larger and more strongly solvated than chloride ions [61]. **With** the addition of 5.0 mM Triclabendazole, both E_{OCP} and E_{PZC} shift marginally, resulting in a reduced excess surface charge of +23 mV. Although the reduction is less pronounced than in HCl, it still confirms the adsorption of Triclabendazole on the mild steel surface [62]. **The** smaller change in surface charge in sulfuric acid may be attributed to competitive adsorption of sulfate ions, which limits the extent of inhibitor adsorption and surface charge neutralization [63].

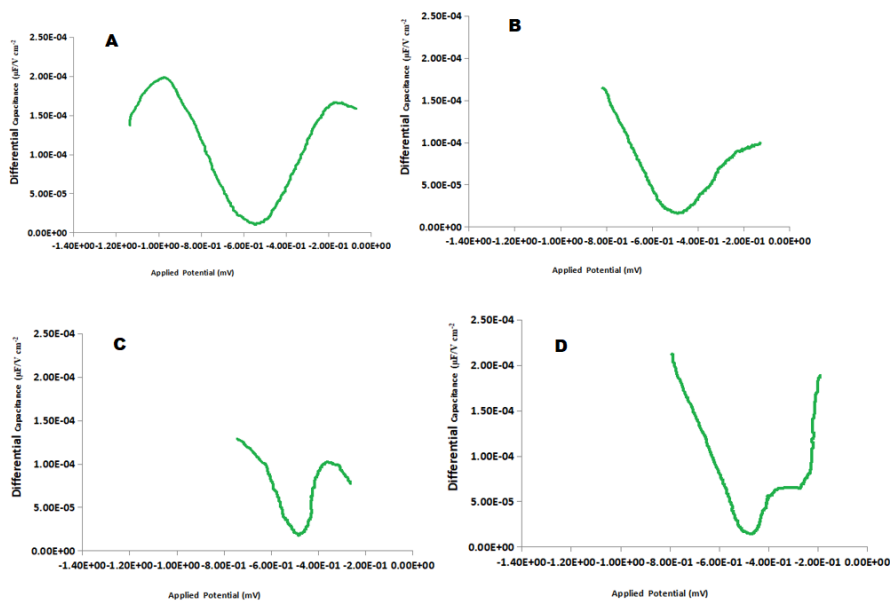


Figure.8 A plot of Applied DC potential vs Differential Capacitance in A. 1.0 M HCl B. 1.0 M HCl + 5.0 mM Triclabendazole C. 0.5 M H₂SO₄ and D. 0.5 M H₂SO₄ + 5.0 mM Triclabendazole

The differential capacitance curves shown in the **Figure 8** reveal a decrease in capacitance values in the presence of Triclabendazole in both acidic media. This **behaviour** is attributed to the replacement of water molecules by inhibitor molecules at the metal–solution interface, leading to an increase in the thickness of the electrical double layer and a decrease in its dielectric constant [55,59]. Such changes in interfacial capacitance further support the adsorption of Triclabendazole and the formation of a compact

protective layer that impedes charge transfer reactions and corrosion processes [60].

The positive excess surface charge observed in both acids indicates that Triclabendazole adsorption is facilitated by electrostatic interactions between protonated inhibitor molecules and the charged metal surface. The greater reduction in surface charge in 1.0 M HCl explains the higher inhibition efficiency observed in this medium, as stronger adsorption leads to better surface coverage and corrosion protection [58,63].

Table.5 Excess surface charge on mild steel surface in 1.0 M HCl and 0.5 M H₂SO₄ in the absence and presence of Triclabendazole

Medium	Inhibitor concentration (mM)	E _{ocp} (mV/SCE)	E _{PZC} (mV/SCE)	Surface Charge
1.0 M HCl	Blank	-490	-532	+42
	5.0	-482	-502	+20
0.5 M H ₂ SO ₄	Blank	-484	-510	+26
	5.0	-489	-512	+23

3.6 Quantum Chemical Studies

Density functional theory (DFT) calculations are widely used to correlate the molecular electronic structure of corrosion inhibitors with their experimental inhibition efficiency. Quantum chemical descriptors such as frontier molecular orbital energies (E_{HOMO} and E_{LUMO}), energy gap (ΔE), electronegativity (χ), global hardness (η), softness (S), electrophilicity index (ω), and fraction of electrons transferred (ΔN) provide valuable insight into adsorption tendency and reactivity of inhibitor molecules on metal surfaces [64,65]. A higher E_{HOMO} value is associated with enhanced electron-donating ability toward vacant d-orbitals of iron, whereas lower E_{LUMO} values indicate improved electron-accepting capability from the metal surface. These

donor–acceptor interactions promote strong adsorption and effective corrosion inhibition [66].

The optimized structure and frontier molecular orbitals of neutral Triclabendazole (**Figures 9 and 10**) show that HOMO electron density is mainly distributed over heteroatoms and aromatic rings, which act as active adsorption centres. This electronic distribution facilitates interaction between the inhibitor molecule and the mild steel surface [67]. The moderate HOMO–LUMO energy gap indicates balanced molecular reactivity and stability, allowing Triclabendazole to adsorb effectively without undergoing rapid degradation. These characteristics explain the appreciable corrosion inhibition efficiency observed experimentally for the neutral molecule [68].

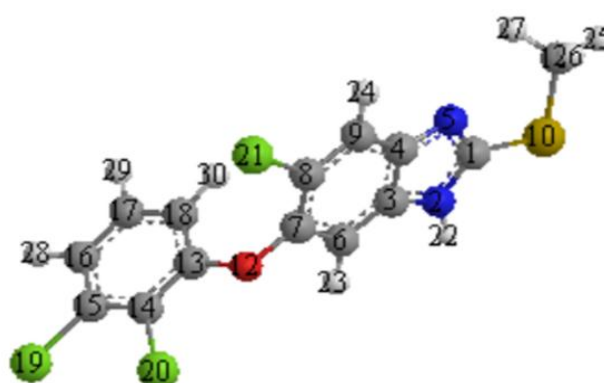


Figure.9 Optimized structure of Triclabendazole

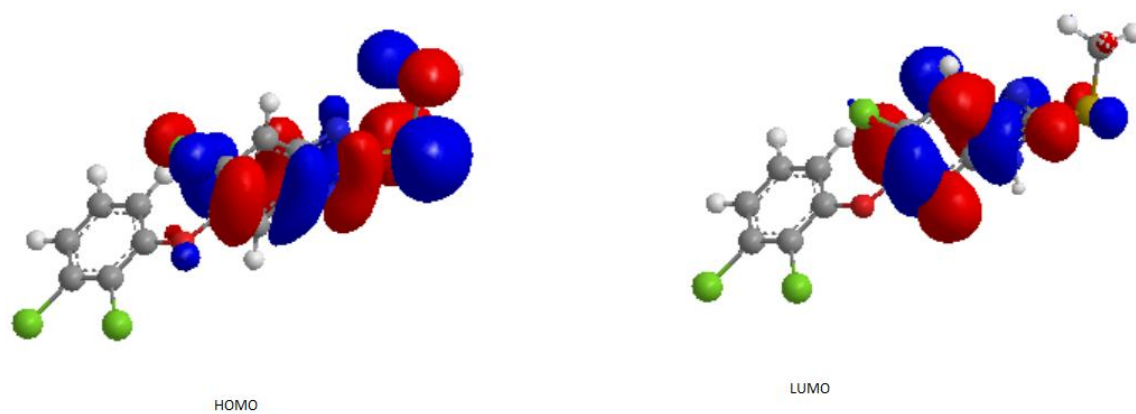


Figure.10 HOMO and LUMO of Triclabendazole

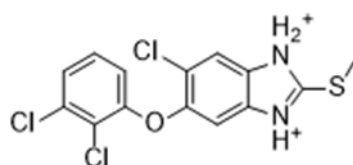


Figure.11 Optimized structure of Triclabendazole protonated at 2N

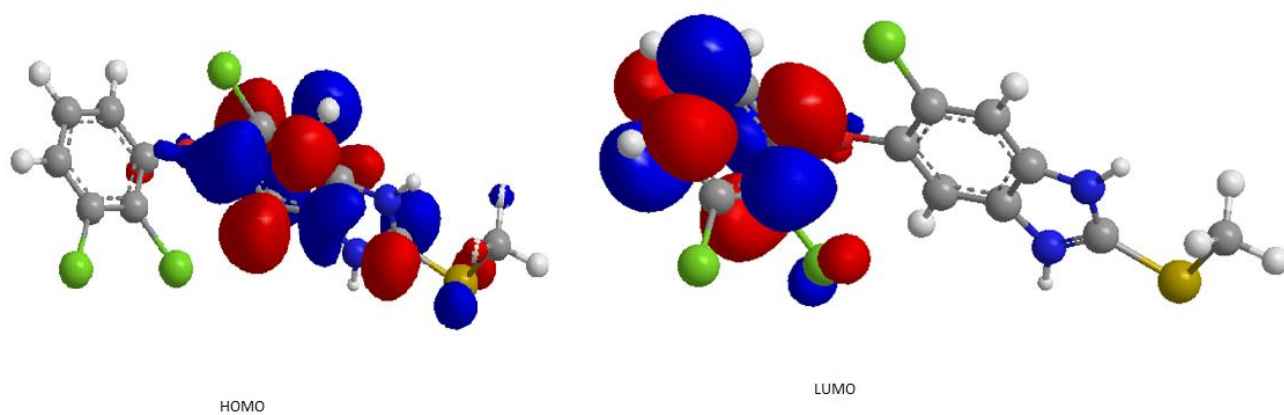


Figure.12 HOMO and LUMO of Triclabendazole protonated at 2N

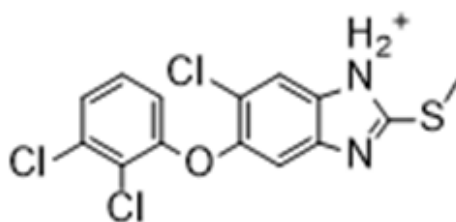


Figure.13 Optimized structure of Triclabendazole protonated at 5N

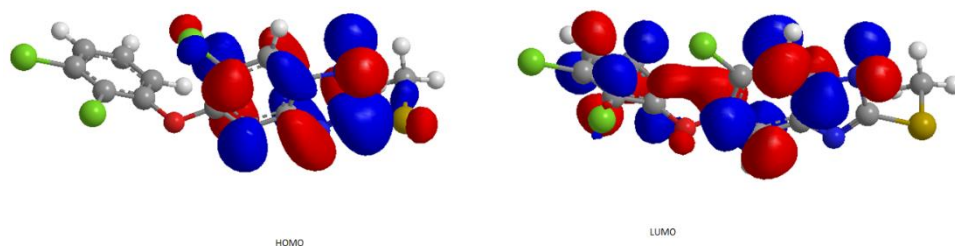


Figure.14 HOMO and LUMO of Triclabendazole protonated at 5N

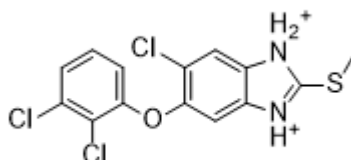


Figure.15 Optimized structure of Triclabendazole protonated at 2N and 5N

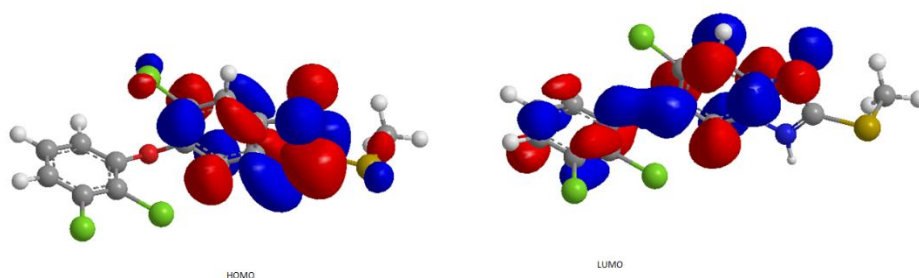


Figure.16 HOMO and LUMO of Triclabendazole protonated at 2N and 5N

In acidic media, Triclabendazole predominantly exists in protonated forms. Protonation at different nitrogen atoms (Figures 11–16) leads to redistribution of electron density and modification of frontier orbital energies. Protonated species generally exhibit:

- Lower E_{LUMO} values, enhancing electron-accepting ability,

- Altered E_{HOMO} levels, affecting electron donation, and
- Increased polarity, which strengthens electrostatic interaction with the charged steel surface [69].

Multiple-site protonation results in reduced ΔE values compared to the neutral form, indicating higher chemical reactivity and stronger adsorption capability [70].

Table.6 Quantum Chemical parameters of inhibitor molecule calculated using DFT

QC Parameters	TRICLABENDAZOLE NEUTRAL	TRICLABENDAZOLE 2N PROTONATED	TRICLABENDAZOLE 6N PROTONATED	TRICLABENDAZOLE 9N PROTONATED	TRICLABENDAZOLE 2 AND 6N PROTONATED	TRICLABENDAZOLE 2 AND 9N PROTONATED	TRICLABENDAZOLE 6 AND 9N PROTONATED	TRICLABENDAZOLE 2, 6 AND 9N PROTONATED
Total energy (eV)	3637.81622	-3650.30974	-3650.30333	-3665.05369	-3664.30891	-3665.05009	-3664.58577	0
E_{LUMO} (eV)	-0.634	-0.224	-0.303	-0.235	-0.465	-0.478	-0.267	-0.504
E_{HOMO} (eV)	-8.558	-8.455	-7.685	-8.298	-8.608	-8.682	-8.256	-7.371
ΔE (eV)	-7.924	-8.231	-7.382	-8.063	-8.143	-8.204	-7.989	-6.867
Ionization potential (I/eV)	0.8555	0.8455244	0.7684504	0.8298168	0.8608064	0.8682497	0.7370855	0.8256039
Electron affinity (A/eV)	8.558	8.455	7.685	8.298	8.608	8.682	8.256	7.371

Hardness (η /eV)	-5.85E-05	0.000122	-0.000248	8.4E-05	3.2E-05	0.0002485	1.95E-05	-7.25E-05
Softness (S/eV)	-8547.008547	4098.360656	-2016.129032	5952.380952	15625	2012.072435	25641.02564	-6896.551724
Electronegativity (χ)	8.5579415	8.455122	7.684752	8.298084	8.608032	8.6822485	8.2560195	7.3709275
Electrophilicity (ω)	-625968.9121	292988.0657	-119063.3333	409870.2266	1157784.608	151672.9155	1747739.948	-374693.6014
Fraction of electrons transferred (ΔN)	-0.29772801	-0.308791029	-0.400962179	-0.326196015	-0.292440969	-0.28473845	-0.444075251	-0.330964597

The HOMO–LUMO energy gap (ΔE) is inversely related to molecular reactivity. Lower ΔE values observed for protonated Triclabendazole species suggest enhanced interaction with the metal surface and improved inhibition efficiency [66]. Similarly, reduced global hardness (η) and increased softness (S) values for protonated species indicate a greater tendency for charge transfer during adsorption. Such behaviour favours the strong inhibitor–metal interactions and stable protective film formation [71]. Electronegativity (χ) values reflect the tendency of inhibitor molecules to attract electrons. The calculated χ values indicate that both neutral and protonated Triclabendazole can effectively participate in back-donation interactions with the iron surface [72].

The electrophilicity index (ω) further demonstrates that protonated species possess a strong ability to stabilize additional electronic charge, enhancing adsorption strength. The negative values of ΔN for all species confirm electron donation from inhibitor molecules to the mild steel surface. Higher magnitude of ΔN for protonated forms indicates stronger donor–acceptor interactions, correlating well with higher corrosion inhibition efficiency [73]. The quantum chemical findings strongly support experimental electrochemical and adsorption results. Enhanced inhibition efficiency of Triclabendazole in acidic media can be attributed to:

- Favourable frontier orbital alignment,
- Reduced energy gap upon protonation,
- Increased softness and charge-transfer capability, and
- Strong adsorption through combined electrostatic and coordinate bonding mechanisms [68–70].

The coexistence of neutral and protonated species in solution ensures effective surface coverage and formation of a compact protective layer on mild steel.

4. CONCLUSION

It was discovered that triclabendazole is a good corrosion inhibitor on mild steel in 1.0 M HCl and 0.5 M H₂SO₄ solutions. Measurements of the rate of weight loss displayed that the corrosion rate significantly decreased with the increasing concentration of the inhibitor and reached its optimal

inhibition efficiencies of approximately 96% in HCl and 93% in H₂SO₄. Electrochemical impedance experiments demonstrated the presence of a protective adsorbed film with increased resistance to charge transfer and reduced double layer capacitance in the presence of Triclabendazole. Potentiodynamic polarization observations demonstrated a significant decrease in the density of corrosion current with a slight change in corrosion potential, and this implies that Triclabendazole is a mixed-type inhibitor.

Adsorption experiments have shown that Triclabendazole adsorption adhered to the Langmuir adsorption isotherm with negative free energy values indicating that Triclabendazole adsorption occurred spontaneously and without fluid exchange interaction involving both physical and chemical interaction. Surface charge assays and differential capacitance assays also supported successful adsorption and surface coverage. Quantum chemistry computation further justified the experimental data, showing good electronic properties and increased reactivity of protonated species in the acidic media, an explanatory point of high corrosion inhibition effectiveness. Overall, Triclabendazole has a great adsorption-controlled corrosion inhibition and an excellent and promising corrosion-inhibitor protonated species in acidic mediums.

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